ANSWER KEY & MARKING SCHEME SET 1

Chapter 10: **Minerals and Energy Resources (Geography)** | Class 10 Social Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** Both A and R are true and R is the correct explanation of A.	Both true; R explains A
2	**(a)** Both A and R are true and R is the correct explanation of A.	Both true; R explains A
3	**(a)** aluminium	Light metal
4	**(a)** Both A and R are true and R is the correct explanation of A.	Both true; R explains A
5	**(a)** Digboi - oil	Oldest oil field
6	**(d)** using public transport	Less fuel use
7	**(a)** Both A and R are true and R is the correct explanation of A.	Both true; R explains A
8	**(d)** nuclear power plants	Atomic power
9	**(c)** A is true but R is false.	A false; R true
10	**(a)** Both A and R are true and R is the correct explanation of A.	Both true; R explains A

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	It gives large energy but has accident and waste risks (1); Safety and cost are debated (1).	1+1
12	Mining displaces tribals and destroys forests (1); It pollutes water and damages livelihoods (1).	1+1
13	Some regions get industry while others lack resources (1); It causes regional inequality and trade dependence (1).	1+1
14	Fossil fuels are limited and polluting (1); Renewable sources are cleaner and technology costs are falling (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Villages need fuel for cooking and light (1); Biogas and solar meet needs (1); This reduces deforestation (1).	1+1+1
16	Coal runs most thermal plants (1); It is cheap and abundant (1); But it pollutes and releases carbon dioxide (1).	1+1+1
17	Solar is available everywhere in sunny regions (1); Wind needs strong steady winds (1); Solar suits scattered villages, wind suits windy coasts (1).	1+1+1
	OR: Government sets policies and prices (1); It supports renewables and imports (1); It promotes saving and efficiency (1).	1+1+1
18	Mining gives raw materials and jobs (1); It damages forests and land (1); Rules, restoration and recycling balance both (1).	1+1+1
	OR: Scrap metal reused cuts mining (1); It saves energy and costs (1); It reduces pollution (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Most metallic minerals lie in the old crystalline rocks of the Peninsular plateau, while coal is found in the Gondwana valleys of the Damodar, Mahanadi, Sone and Godavari (1); Iron ore belts are in Odisha-Jharkhand, Chhattisgarh and Karnataka; Odisha leads in manganese and bauxite, and Madhya Pradesh in copper (1); Petroleum comes from Mumbai High, Gujarat and Assam, while limestone and other minerals occur in sedimentary rocks of the western and southern parts (1); This uneven distribution gives mineral-rich states raw materials for steel, aluminium and power industries and export earnings (1); Yet many of these states remain poor and suffer environmental damage, so benefits must be shared more fairly (1).	1+1+1+1+1

	OR: Solar energy has huge potential in India because of abundant sunshine and can be used in remote villages through photovoltaic cells (1); Wind energy is developed in Tamil Nadu, Gujarat, Karnataka and Andhra Pradesh (1); Biogas uses dung and farm waste for cooking and gives manure (1); Limits include high setup costs, dependence on weather, storage problems and need for suitable sites (1); Research, investment and policy support can overcome them (1).	1+1+1+1+1
20	Thermal power is the largest source, using coal, petroleum and natural gas, and plants are located near coalfields (1); Hydroelectricity uses falling water, is renewable and clean, but needs dams and suitable relief, as at Bhakra-Nangal (1); Nuclear power uses uranium and thorium, and stations are at Tarapur, Kota and Kalpakkam (1); Thermal power causes pollution, hydel power displaces people, and nuclear power carries safety and waste risks (1); A balanced mix is necessary to meet rising demand (1).	1+1+1+1+1
	OR: Coal is the most abundant fossil fuel in India and is the main source of thermal power, but it pollutes the air (1); Petroleum from Mumbai High, Gujarat and Assam and natural gas from the Krishna-Godavari basin supply fuel, CNG and raw material for industries, but India imports much crude oil (1); Hydroelectricity, nuclear power from uranium and thorium, and non-conventional sources like solar, wind, biogas, tidal and geothermal are being developed (1); India has abundant sunshine, and the largest cluster of wind farms stretches from Nagercoil to Madurai (1); A shift to renewable sources, along with energy conservation, is required for sustainable supply (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

ANSWER KEY & MARKING SCHEME SET 2

Chapter 10: **Minerals and Energy Resources (Geography)** | Class 10 Social Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** biogas	Organic waste is used
2	**(c)** Mumbai High - coal	Mumbai High is an oilfield
3	**(a)** non-conventional energy sources are eco-friendly	Solar and wind are clean
4	**(c)** it is dangerous	Narrow tunnels
5	**(d)** making steel	Mica is an insulator
6	**(a)** Both A and R are true and R is the correct explanation of A.	Both true; R explains A
7	**(b)** India's import bill	India imports crude oil
8	**(a)** Both A and R are true and R is the correct explanation of A.	Both true; R explains A
9	**(d)** A is false but R is true.	A false; R true
10	**(a)** peat, lignite, bituminous, anthracite	Anthracite is best

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	CNG burns cleaner than petrol or diesel (1); It reduces air pollution in cities (1).	1+1
12	Collecting and storing solar energy is costly (1); Dependence on weather and land needs limit scale (1).	1+1
13	Mining scars the land (1); Restoring land protects environment and livelihoods (1).	1+1
14	Exporting raw minerals depletes limited stocks (1); Value is gained by others if not processed at home (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Domestic crude oil is far less than demand (1); Transport and industry demand has grown (1); So India imports much oil (1).	1+1+1
16	Technology is costly and new (1); Research makes it efficient (1); Investment expands use (1).	1+1+1
17	Imported crude oil arrives by sea (1); Ports reduce transport costs (1); Refined products can be shipped out (1).	1+1+1
	OR: Better methods extract more from ore (1); Waste can be recycled (1); Low-grade ores become usable (1).	1+1+1
18	Coal burning releases carbon dioxide and ash (1); It causes air pollution and global warming (1); Mining also damages land (1).	1+1+1
	OR: Develop solar, wind and biogas (1); Improve efficiency (1); Use public transport (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Minerals are finite and non-renewable, taking millions of years to form, while rich deposits are few and costly to extract as they deplete (1); Judicious use avoids wastage during mining and processing (1); Recycling of metals such as scrap iron, copper and aluminium reduces the demand for fresh ore (1); Substitutes and better technology can reduce use of scarce minerals (1); Strict regulation of mining and restoration of mined land also help to preserve minerals and the environment for future generations (1).	1+1+1+1+1
	OR: Mining clears forests and leaves deep scars on land, and the rat-hole mines of Meghalaya are narrow, unsafe tunnels that collapse or flood (1); Dust and toxic waste pollute air and water, and mining waste and effluents pollute rivers (1); Miners suffer from lung diseases and accidents (1); Tribal communities living in mining areas lose land and livelihood and may be displaced (1); Strict safety rules, restoration of mined land and fair compensation are needed (1).	1+1+1+1+1

20	Conventional sources such as coal, petroleum, natural gas and hydel power are widely used, but fossil fuels are exhaustible and pollute (1); Non-conventional sources such as solar, wind, tidal, geothermal and biogas are renewable and eco-friendly (1); Conventional sources have established technology and large capacity, while non-conventional ones are costlier to set up and site-specific (1); Examples: thermal plants near coalfields versus wind farms in Tamil Nadu and a solar plant at Bhuj (1); India needs both in the present but must increase non-conventional sources in the long run (1).	1+1+1+1+1
	OR: Citizens can save electricity by switching off unused lights and using efficient appliances (1); They can use public transport and avoid wasteful use of fuel (1); They can use non-conventional sources such as solar heaters and biogas where possible (1); They can avoid wasting metal products and reuse or recycle scrap (1); Everyday small steps by many people reduce pressure on minerals and energy resources (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

ANSWER KEY & MARKING SCHEME SET 3

Chapter 10: **Minerals and Energy Resources (Geography)** | Class 10 Social Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** the Gulf of Khambhat and Gulf of Kuchchh	Coastal inlets
2	**(b)** Japan and South Korea	Through ports
3	**(b)** thermal power	Coal-based power
4	**(b)** switching off unused lights	Reduces demand
5	**(a)** it needs strong and steady winds	Site specific
6	**(a)** coal	A fossil fuel
7	**(a)** Both A and R are true and R is the correct explanation of A.	Both true; R explains A
8	**(a)** thermal power generation	Brown coal
9	**(b)** location advantage	Cheap fuel
10	**(a)** reduce the cost of transporting bulky ore	Raw material oriented

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Dams displace people and submerge forests (1); They change river flow and cause conflicts (1).	1+1
12	Factories stop or slow production (1); Losses and job cuts follow (1).	1+1
13	Saving energy reduces demand and imports (1); It conserves resources and the environment (1).	1+1
14	They are inexhaustible and cause less pollution (1); They reduce dependence on fossil fuel imports (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Tidal energy needs special coastal sites (1); Geothermal needs hot rock zones (1); Both are costly and limited in India (1).	1+1+1
16	Mumbai High is offshore and the largest producer (1); Digboi in Assam is the oldest onshore field (1); Mumbai High needs costly rigs (1).	1+1+1
17	Mineral-rich areas get industries and jobs (1); But some, like Jharkhand, remain poor (1); Local people need fair benefits (1).	1+1+1
	OR: Nuclear plants have strict safety systems (1); But accidents and waste disposal are serious risks (1); Safety needs constant care (1).	1+1+1
18	Industries use iron, copper, aluminium, limestone and fuel (1); Without minerals, no steel, machinery or cement (1); So minerals support industrial growth (1).	1+1+1
	OR: More vehicles raise fuel demand (1); Road transport depends on oil (1); Public transport and CNG reduce use (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Government regulates mining through rules and leases and promotes safety and restoration (1); It sets prices and subsidies for fuels and electricity (1); It promotes renewable energy and energy-saving devices (1); It runs public sector undertakings in coal, oil and power (1); Policy shapes the sustainable use of minerals and energy (1).	1+1+1+1+1
	OR: Steel and cement plants are located near iron ore, coal and limestone because these raw materials are bulky to transport (1); Thermal power plants are built near coalfields to save transport cost (1); Aluminium smelters need cheap and constant electricity, so they are located near power supply (1); Hydel stations attract industries to hilly river regions (1); Availability of minerals and power therefore decides where industries grow (1).	1+1+1+1+1

20	India depends heavily on imported crude oil, so oil price rises affect the economy (1); Demand for electricity and fuel is rising rapidly with industrialisation and transport (1); Coal is abundant but polluting, and hydel and nuclear power have limits (1); Renewable sources need investment and technology (1); Security requires diversification of sources, conservation and efficient use (1).	1+1+1+1+1
	OR: Coal is concentrated in the Gondwana fields of eastern India, so thermal power and steel industries cluster there (1); Oil comes from Mumbai High, Assam and Gujarat, with refineries on the coast and in Assam (1); Large oil imports shape the economy and the balance of payments (1); Natural gas pipelines like the Hazira-Vijaipur-Jagdishpur line link fields to fertiliser and power plants (1); The geography therefore influences industrial location, trade and regional development (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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